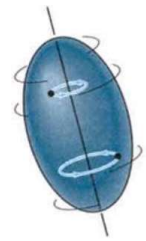
Movement of Translation: it is a movement in which **all the points of the solid have at each moment the same velocity and the same acceleration vectors**. In addition, **any segment of the solid remains parallel to itself during the movement**.

The translational movement can be either rectilinear (straight-line path) or curvilinear (curved path).

If the curved line is a circle or an arc of a circle, the translation is circular.



Rotation about a fixed axis

Rotational movement about a fixed axis: it is a movement during **which all the points of the solid describe circular trajectories having their centers on the axis**, and traced in planes perpendicular to that axis. **All the points of the solid have the same angular speed and acceleration, at the same instant.**

Points closer to the center have smaller linear velocities and further points have larger linear velocities.

For a combined movement, the body performs both types, translation and rotation at the same time.

When you are interested in pushing a car, holding an object in place, or compressing a spring, you often talk about **force**.

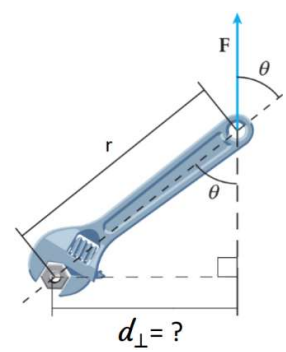
However, when opening a door, balancing the arms of a scale, or loading a spiral spring, we often talk about the **moment of force**.

The rotational effect of a force is called "**moment of the force**".

A force $\vec{F}$ **parallel to the axis of rotation** has no effect of rotation around this axis.

A force $\vec{F}$ **whose line of action passes through the axis of rotation** does not have a rotation effect around this axis.

For a force $\vec{F}$ exerted on a moving solid about an axis, to have a rotation effect **(moment)** it is necessary that $\vec{F}$ is **<u>neither parallel to</u>**, **<u>nor meeting the axis</u>**.



Experience shows that the effect of rotation, or the moment of a force about an axis, depends on **the magnitude of this force, its distance to the axis, and its direction:**

$$\boxed{\mathfrak{M}_{\vec{F}/\Delta} = F.d_\perp = F.\,r.sin(\theta)}$$

"r" is the length of the turning radius and θ is the angle between this radius and the applied force.

Moment is an algebraic quantity; we count (+) the moment of a force, that tends to turn the solid in the (+) direction chosen. Meanwhile, the moment is (-) if this force tends to turn the body in the direction (-):

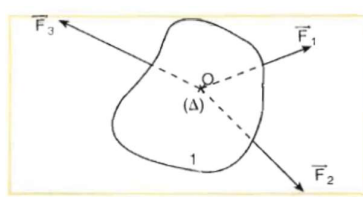
$$\boxed{\mathfrak{M}_{\vec{F}/\Delta} = \pm F.d_\perp}$$

The resultant moment $\mathfrak{M}_{tot/\Delta}$ of several forces applied to a solid is the algebraic sum of the moments of these forces with respect to (Δ).

$\mathfrak{M}_{tot/\Delta} > 0 \Rightarrow$ the solid turns in the (+) direction chosen. However, $\mathfrak{M}_{tot/\Delta} < 0 \Rightarrow$ solid turns in the direction (-).

In static equilibrium, the object is:

- neither in translational movement⇒ $\boxed{\sum \vec{F}_{ext/solid} = \vec{0}}$
- nor in rotation⇒ $\boxed{\sum \mathfrak{M}_{\vec{F}_{ext}/\Delta} = 0}$

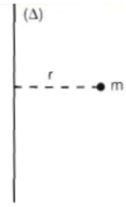


In the case of a solid subjected to 3 non-parallel forces, these 3 forces are concurrent and coplanar.

The mass **m** of a body is a measure of the **translational inertia** of this body. Inertia is a resistance of the body to acceleration when applying a force $\vec{F}$. A body tends not to accelerate (or change its velocity $\vec{V}$).

Similarly, *two solids*, movable about a fixed axis (Δ), and submitted to the same force, *do not necessarily undergo the same rotational motion about* (Δ).

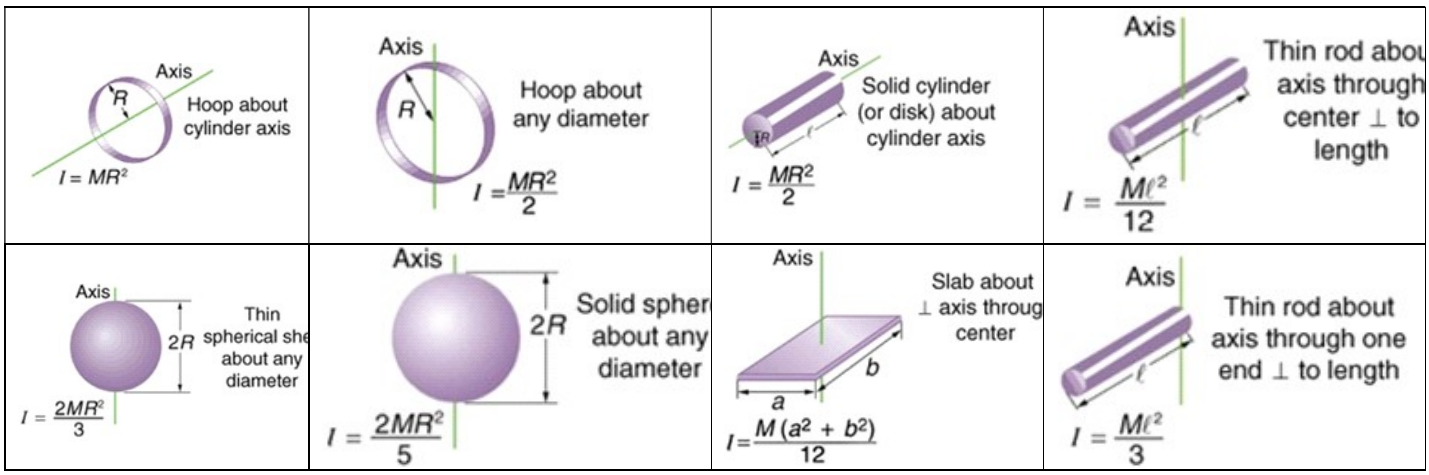
The inertia of rotation of a solid with respect to an axis (Δ) is measured by a physical quantity called the **moment of inertia**.



The moment of inertia, with respect to an axis (Δ), of a particle, of mass m, and situated at a distance r from (Δ) is: $\boxed{I_\Delta = m \cdot r^2}$

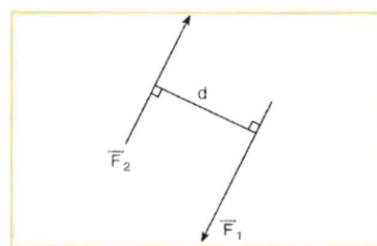
In S.I. units, m is in (kg), r in (m), then I is in (kg.m^2).

The moment of inertia of a system of particles P_1, P_2, P_3, …. , about an axis (Δ) is : $\boxed{I_\Delta = \sum m_i \cdot r_i^2}$
Where m_i is the mass of particle P_i and r_i its distance to (Δ).



Moment of inertia of some solids about a given axis

We call couple of forces, the set of 2 forces which: have the same direction, are not on the same line, have 2 opposite directions, have the same magnitude.



If the plane of forces is perpendicular to the axis of rotation, the couple moment is independent of the position of the axis of rotation. This moment is equal to the product of the common intensity of the two forces (F1 = F2 = F) by the distance "d " between the two forces: $\boxed{\mathfrak{M}_{Couple} = F.d}$

For a solid rotating around a fixed axis (Δ), Newton's 2nd law can be written:

$$\boxed{\sum \mathfrak{M}_{\vec{F}_{ext\ on\ solid\ about\ (\Delta)}} = I.\theta''}$$

With: I is the moment of inertia of the solid about (Δ) and θ" is the solid angular acceleration.

If the resulting moment on a solid is constant then the angular acceleration is constant:

By antiderivative	Angular acceleration	$\theta'' = \frac{\mathfrak{M}_{tot}}{I} = constant$	
By antiderivative	Angular velocity	$\theta' = \theta'' \cdot t + \theta'_0$	time eliminated between θ and θ'
By antiderivative	Angular abscissa	$\theta = \frac{1}{2}\theta'' \cdot t^2 + \theta'_0 \cdot t + \theta_0$	$(\theta')^2 - (\theta'_0)^2 = 2 \cdot \theta" \cdot (\theta - \theta_0)$